4 Uniform $(0,\theta)$ distribution

$X_1, X_2, \ldots, X_n \sim U(0,\theta)$.

$$f(x_i;\theta) = \begin{cases} 1/\theta & \text{if } 0 \le x_i \le \theta \\ 0 & \text{otherwise} \end{cases}$$

$$f(\{x_i\}^n;\theta) = \begin{cases} 1/\theta^n & \text{if } \forall i,\ 0 \le x_i \le \theta \\ 0 & \text{otherwise} \end{cases}$$

so that we maximize the likelihood

The ML est. of θ must be the smallest number that is $\geq \max_i \{X_i\}$

$\hookrightarrow \hat{\theta} = \max_i \{X_i\}$

4.1 Unif (v,θ) $v < \theta$

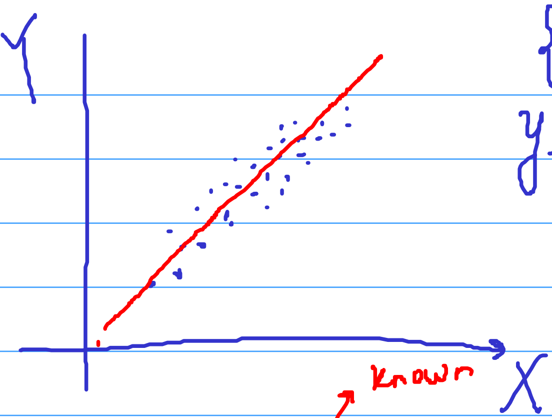
$$f(x_i;\theta, v) = \begin{cases} \frac{1}{\theta - v} & \text{if } v \le x_i \le \theta \\ 0 & \text{otherwise} \end{cases}$$

$$f(\{x_i\}_{i=1}^n;\theta, v) = \begin{cases} \frac{1}{(\theta - v)^n} & \text{if } \forall i,\ v \le x_i \le \theta \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{\theta} = \max_i \{X_i\}$$

$$\hat{v} = \min_i \{X_i\}$$

$\hookrightarrow$ we want the largest possible number that is $\le$ all the x_i values



noiseless, noisy

$$\{(x_i, y_i)\}_{i=1}^{n}$$

$$y_i = \underline{mx_i + c} + \varepsilon_i$$

$$\underline{\varepsilon_i \sim N(0, \sigma^2)}$$

known, unknown

$$y_i \sim N(mx_i + c, \sigma^2)$$

$$p(y_i ; x_i, m, c) = \frac{e^{-(y_i - mx_i - c)^2/2\sigma^2}}{\sigma\sqrt{2\pi}}$$

$$p(\{y_i\}; \{x_i\}, m, c) = \prod_{i=1}^{n} \frac{e^{-(y_i - mx_i - c)^2/2\sigma^2}}{\sigma\sqrt{2\pi}}$$

$$\log p = \sum_{i=1}^{n}\left[-\frac{(y_i - mx_i - c)^2}{2\sigma^2}\right] - n\log\sigma - \frac{n}{2}\log(2\pi)$$

$$\frac{\partial \log p}{\partial m} = \sum_{i=1}^{n} \frac{+\cancel{2}(y_i - mx_i - c)(+x_i)}{\cancel{2}\cancel{\sigma^2}} = 0$$

$$m\sum_{i=1}^{n} x_i^2 + c\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i \longrightarrow 1$$

$$\frac{\partial \log p}{\partial c} = \sum_{i=1}^{n} \frac{+\cancel{2}(y_i - mx_i - c)(+1)}{\cancel{2}\cancel{\sigma^2}} = 0$$

$$m\sum_{i=1}^{n} x_i + cn = \sum_{i=1}^{n} y_i \longrightarrow 2$$

$$\hat{c} = \sum_{i=1}^{n} \frac{y_i}{n} - \hat{m}\sum_{i=1}^{n} \frac{x_i}{n} \longrightarrow A$$

$$\hat{m} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \qquad \bar{x} = \sum_{i=1}^{n} x_i/n \qquad \bar{y} = \sum_{i=1}^{n} y_i/n$$

$$y_i = mx_i + c + \varepsilon_i$$

$$\underset{n\times 1}{\vec{y}} = \underset{n\times 2}{\left(\overset{n\times 1}{\vec{x}} \;\middle|\; \overset{n\times 1}{\vec{1}} \right)} \underset{2\times 1}{\begin{pmatrix} m \\ c \end{pmatrix}}$$

$$= \underset{n\times 2}{X} \begin{pmatrix} m \\ c \end{pmatrix} \rightarrow \vec{\theta} \text{ (2x1 vector)}$$

$$X^T \vec{y} = X^T X \vec{\theta}$$

$$\hat{\vec{\theta}} = (X^T X)^{-1} X^T \vec{y}$$

$\rightarrow$ minimize of $\|\vec{y} - X\vec{\theta}\|_2^2$

$$Z_i = ax_i + by_i + c + \varepsilon_i$$

$$\|\vec{y} - X\vec{\theta}\|_2^2 = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

der. of $\|\vec{y} - X\vec{\theta}\|^2$ w.r.t. $\vec{\theta}$ and set it to $\vec{0}$

$$(-X^T)\cancel{2}(y - X\theta) = 0$$

$$X^T y = X^T X\theta$$

$$\sum_{i=1}^{n} (y_i - (X\theta)_i)^2 = \sum_{i=1}^{n} \Big(y_i - \sum_j X_{ij}\theta_j\Big)^2$$

$$z_i = a x_i + b y_i + c + \varepsilon_i$$